Donglai Yang

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Earth & Atmospheric Sciences ⋄ Georgia Institute of Technology

Biography

Donglai Yang is a 3rd-year Ph.D. candidate in Polar Geophysical Simulation Lab at Georgia Institute of Technology and a Polar Informatics Grad Fellow at iHARP. He is broadly interested in ice sheet dynamic change prediction with numerical modeling, ice-penetrating radar observations, and ML/AI-empowered Bayesian inverse problem to bridge multi-scale geophysical observations and modeling insights.

Education

Georgia Institute of Technology, Georgia — 08/2023 - Present
Ph.D. Candidate of Geophysics, Earth and Atmospheric Sciences — *Advisor: Winnie Chu*

University at Buffalo, New York — 08/2021 - 05/2023
M.S. Computational Geosciences — *Advisor: Kristin Poinar, Sophie Nowicki*

Wesleyan University, Connecticut — 08/2017 - 06/2021
B.A. Physics; B.A. Earth Sciences — *Thesis advisor: Phillip Resor*

Journal Publications

1. **Yang D.**, Chu W., Dawson E., 2025 (In preparation) *Ice Sheet Basal Temperature Inference by Bridging Airborne Radar and Thermal Modeling with Generative Deep Learning*. Planned submission: Journal of Geophysical Research - Machine Learning and Computation.
2. Dawson E., Wolfenbarger N., Feduschak M., Chu W., **Yang D.**, Schroeder D. 2025 (In revision) *Radar attenuation through ice on Earth and across the solar system*. Review of Geophysics.
3. Dawson E., Chu W., Christoffersen M., **Yang D.**, Farris S., MacGregor J. 2026 *Ice sheet attenuation from radar sounding in the frequency domain*. J. Glaciol. doi.org/10.1017/jog.2026.10130
4. **Yang D.**, Poinar K., Csatho B., Narkevic A., Gao H., Chu W., 2025 (In review) *Ice dynamic thinning observation and modeling reveal decadal transition in the grounding state of Helheim Glacier from 1998 to 2022*. Geophysical Review Letter.
5. **Yang D.**, Poinar K., Nowicki S., Csatho B., 2024. *Characteristics of dynamic thickness change across diverse outlet glacier geometries and basal conditions*. Journal of Glaciology. Published online 2024:1-15. doi:10.1017/jog.2024.50

Presentations

† represents oral presentations; ∗ represents poster presentations; • represents student mentees; no symbol represents presentations by collaborators

1. † **Yang D.**, Chu W., Dawson E., Morlighem M., 2026. *A Bayesian Framework for Mapping Ice Sheet Basal Temperature*. Invited talk at IGS ECR Global Seminar.
2. • Liu L., **Yang D.**, Chu W., MacGregor J., 2025. *Using Thermal Model Ensemble and Machine Learning to Predict the Basal Temperature of the Northern Greenland Ice Sheet*. AGU 2025 Fall Meeting conference.
3. • Scalabrin J., **Yang D.**, Chu W., 2025. *Reconstructing Ross Ice Shelf in 1970's using Archival Geophysical Data and Physics-Informed Machine Learning*. AGU 2025 Fall Meeting conference.
4. † **Yang D.**, Chu W., Dawson E., 2025. *Probabilistic Inference of Ice Sheet Basal Temperature with Thermal Modelling and Radar Attenuation*. EGU General Assembly 2025.
5. † **Yang D.**, Chu W., Dawson E., 2024. *Probabilistic Ice Basal Temperature Inversion with Ice-Sheet Model and Generative AI*. AGU 2024 Fall Meeting conference.
6. Mejia J., Poinar K., Meyer C., Chu W., **Yang D.**, 2023. *Investigating Observations and Modeling to investigate Firn Aquifer Hydrology and its Role in Crevasse Propagation on Helheim Glacier*. AGU 2023 Fall Meeting conference.
7. † **Yang D.**, Poinar K., Nowicki S., 2022. *Synthetic glacier experiments reveal controls on ice dynamic thinning variability*. AGU 2022 Fall Meeting conference.
8. † **Yang D.**, Poinar K., Nowicki S., 2022. *Modeling characteristic surface elevation changes reveal dominant process control*. NASA ICESat-2 Science Meeting.
9. ∗ **Yang D.**, Resor P., 2022. *Numerical Modeling of Frictional Melting Dynamics Constrained by Surface Micro-Roughness*. EGU General Assembly 2021.

Honors & Fellowships

2nd place in poster presentation - Georgia Tech EAS Graduate Student Research Symposium – 2026
1st place in oral presentation - Georgia Tech EAS Graduate Student Research Symposium – 2025
NSF iHARP Polar Informatics Graduate Fellow (2024-2025) – 2024
1st place in poster presentation - Georgia Tech EAS Graduate Student Research Symposium – 2024
Phi Beta Kappa National Honor Society– 2021
High Honor in Earth & Environmental Sciences – 2021

Teaching Experience

Georgia Institute of Technology
Graduate Teaching Assistant, Earth Processes Lab *EAS2600*
Undergraduate-level introduction to physical processes on Earth

University at Buffalo
Graduate Teaching Assistant, Natural Hazards and Climate Change *GLY105*
Undergraduate-level introduction to climate change, geology, and hazards

Mentorship Experience

Leo Liu (Emory U., Physics; 2025-present)
Sophomore undergraduate: inference of basal temperature in Northern Greenland Ice Sheet

Jacob Scalabrin (Florida State U., Scientific Computing; 2025-present)
Senior undergraduate: reconstruction of Ross Ice Shelf in 1970s with Physics-Informed Machine Learning

Leadership Experience

Executive Vice President
Graduate Student Government Association (GSGA, 2024-2025) at Georgia Tech
Assisted and promoted student campus-wide events and facilitates two-way conversations between graduate student body and the university administration. Initiated a campus-wide peer mentorship program for international students "Buzz Buddies" attracting ~ 250 *participants.*

Vice President of Student Life
Graduate Student Government Association (GSGA, 2025-2026) at Georgia Tech
Spearheading the peer mentorship program Buzz Buddies and engaging with university administrators to broaden graduate students' participation in on-campus activities; chairing Graduate International Student Affairs committee.

Work Experience

Sust Global
Climate Data Science Intern, 06/2022 - 08/2022
Engineered model-based future physical climate risk index for heatwave, extreme precipitation, and sea level rise.

Workshop and Field Experience

Connected Data Products Hackdays (2026)
Participated in the 3-day Hackathon on open data science in Cryospheric application. Proposed and coordinated the prototyping of small Python package on connecting the outputs of two open-source python packages (icepyx and xopr) to faciliate glaciological interpretation.

Patagonian Icefield Research Program (2024)
Participated in the 10-day expedition at Bernal Glacier in Patagonia, Chile. Assisted with basic logistic support and trailblazing. Received various field technique trainings including GPS, radar, drone photogrammetry. Conducted ablation stake installation and measurement. Presented preliminary results to the local community in Puerto Natales.

GlaMacLeS: The Glaciology in Machine Learning Summer School (2024)
Participated in the one-week workshop on deep-learning application to glaciological problems. Completed a small project with Dr.Mauro Perego on using operator learning (DeepONet) to emulate ice sheet models.

Academic Services

Reviewer: Papers: Journal of Glaciology

University committees: Student Grievance and Appeal committee (Fall 2024 – Spring 2025) ⋄ College of Lifetime Learning Dean Search committee (Fall 2024 – Spring 2025)

Community committee: Canvassing Committee (2026), AGU Cryosphere Section.